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LAB ASSIGNMENT (1)

Objective:

- To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create a new database named DBMS_YourName.
3. Draw ER diagrams for the given scenarios.
4. Convert entities and relationships into relational schemas.
5. Create tables using the derived schemas.

Relational Schema

PART A – COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)

PART C – CUSTOMER SALE SCHEMA

- Database Schema for a customer-sale scenario
- Customer(Cust id : integer, cust_name: string)
- Item(item_id: integer, item_name:string, price: integer)
  Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

```
CREATE DATABASE DBMS_Vanshika;
USE DBMS_Vanshika;

CREATE TABLE STUDENT (
   USN VARCHAR(20) PRIMARY KEY,
   SName VARCHAR(50),
   Address VARCHAR(100),
   Phone VARCHAR(15),
   Gender VARCHAR(10)
);

CREATE TABLE SEMSEC (
   SSID VARCHAR(10) PRIMARY KEY,
   Sem INT,
   Sec VARCHAR(5)
);

CREATE TABLE CLASS (
   USN VARCHAR(20),
   SSID VARCHAR(10),
   PRIMARY KEY (USN, SSID),
   FOREIGN KEY (USN) REFERENCES STUDENT(USN),
   FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

CREATE TABLE SUBJECT (
   Subcode VARCHAR(10) PRIMARY KEY,
   Title VARCHAR(50),
   Sem INT,
```

```
    Credits INT
);

CREATE TABLE IAMARKS (
    USN VARCHAR(20),
    Subcode VARCHAR(10),
    SSID VARCHAR(10),
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

CREATE TABLE EMPLOYEE (
    SSN VARCHAR(20) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex VARCHAR(10),
    Salary INT NOT NULL,
    Superssn VARCHAR(20),
    Dno INT,
    FOREIGN KEY (Superssn) REFERENCES EMPLOYEE(SSN)
);

CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
```

```
    Mgrssn VARCHAR(20),
    Mgrstartdate DATE,
    FOREIGN KEY (Mgrssn) REFERENCES EMPLOYEE(SSN)
);

CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);

CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50),
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);

CREATE TABLE WORKS_ON (
    SSN VARCHAR(20),
    PNo INT,
    Hours INT,
    PRIMARY KEY (SSN, PNo),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);

CREATE TABLE CUSTOMER (
    Cust_id INT PRIMARY KEY,
```

```
    cust_name VARCHAR(50)
);

CREATE TABLE ITEM (
    item_id INT PRIMARY KEY,
    item_name VARCHAR(50),
    price INT NOT NULL
);

CREATE TABLE SALE (
    bill_no INT PRIMARY KEY,
    bill_date DATE,
    cust_id INT,
    item_id INT,
    qty_sold INT,
    FOREIGN KEY (cust_id) REFERENCES CUSTOMER(Cust_id),
    FOREIGN KEY (item_id) REFERENCES ITEM(item_id)
);

ALTER TABLE STUDENT
ADD Email VARCHAR(100);

ALTER TABLE EMPLOYEE
MODIFY Salary INT NOT NULL;

ALTER TABLE ITEM
MODIFY price INT NOT NULL;

ALTER TABLE STUDENT
RENAME COLUMN SName TO StudentName;
```

```
DROP TABLE DLOCATION;

INSERT INTO CUSTOMER VALUES
(1,'Vanshika'),
(2,'Roshani'),
(3,'Sneha'),
(4,'Priya'),
(5,'Karan'),
(6,'Meena'),
(7,'Arjun'),
(8,'Divya'),
(9,'Rohit'),
(10,'Anjali');

INSERT INTO ITEM VALUES
(101,'Pen',20),
(102,'Notebook',50),
(103,'Bag',800),
(104,'Shoes',1200),
(105,'Watch',1500),
(106,'Bottle',150),
(107,'Calculator',600),
(108,'Mouse',400),
(109,'Keyboard',700),
(110,'Book',300);

INSERT INTO SALE VALUES
(1,CURDATE(),1,101,5),
(2,CURDATE(),2,104,1),
(3,CURDATE(),3,105,2),
(4,'2026-02-25',4,103,1),
```

(5,'2026-02-24',5,110,3),

(6,'2026-02-23',6,107,1),

(7,'2026-02-22',7,108,2),

(8,'2026-02-21',8,102,4),

(9,'2026-02-20',9,109,1),

(10,'2026-02-19',10,106,5);

```
SELECT s.bill_no, c.cust_name, i.item_name
FROM SALE s
JOIN CUSTOMER c ON s.cust_id = c.Cust_id
JOIN ITEM i ON s.item_id = i.item_id
WHERE s.bill_date = CURDATE();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_no	cust_name	item_name
1	Vanshika	Pen
2	Roshani	Shoes
3	Sneha	Watch

```
SELECT s.bill_no, (i.price * s.qty_sold) AS Total_Amount
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_no	Total_Amount
1	100
2	1200
3	3000
4	800
5	900
6	600
7	800
8	200
9	700
10	750

Result 3

```
SELECT DISTINCT c.cust_name
FROM SALE s
JOIN CUSTOMER c ON s.cust_id = c.Cust_id
JOIN ITEM i ON s.item_id = i.item_id
WHERE i.price > 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

cust_name
Priya
Roshani
Sneha
Meena
Arjun
Rohit
Karan

```
SELECT c.cust_name, COUNT(s.item_id) AS Total_Products
FROM SALE s
JOIN CUSTOMER c ON s.cust_id = c.Cust_id
GROUP BY c.cust_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

cust_name	Total_Products
Vanshika	1
Roshani	1
Sneha	1
Priya	1
Karan	1
Meena	1
Arjun	1
Divya	1
Rohit	1
Anjali	1

```
SELECT i.item_name
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id
WHERE s.cust_id = 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_name
Book

```
CREATE VIEW Bill_View AS
SELECT s.bill_no, s.bill_date, s.cust_id, s.item_id,
    i.price, s.qty_sold,
    (i.price * s.qty_sold) AS Amount
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id;
```

```
CREATE VIEW Weekly_Sales AS
SELECT bill_date, SUM(i.price * s.qty_sold) AS Daily_Total
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id
WHERE bill_date >= CURDATE() - INTERVAL 7 DAY
```

GROUP BY bill_date;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	18:33:15	SHOW DATABASES	9 row(s) returned	0.016 sec / 0.000 sec
7	18:36:23	DROP DATABASE DBMS_Vanshika	5 row(s) affected	0.157 sec
8	18:37:01	CREATE DATABASE DBMS_Vanshika	1 row(s) affected	0.032 sec
9	18:37:01	USE DBMS_Vanshika	0 row(s) affected	0.000 sec
10	18:59:49	CREATE TABLE STUDENT (USN VARCHAR(20) PRIMARY KEY, SName VARCHAR(50), Address V...	0 row(s) affected	0.063 sec
11	18:59:54	CREATE TABLE SEMSEC (SSID VARCHAR(10) PRIMARY KEY, Sem INT, Sec VARCHAR(5))	0 row(s) affected	0.047 sec
12	18:59:59	CREATE TABLE CLASS (USN VARCHAR(20), SSID VARCHAR(10), PRIMARY KEY (USN, SSID), ...	0 row(s) affected	0.062 sec
13	19:00:07	CREATE TABLE SUBJECT (Subcode VARCHAR(10) PRIMARY KEY, Title VARCHAR(50), Sem INT, ...	0 row(s) affected	0.047 sec
14	19:00:14	CREATE TABLE IAMARKS (USN VARCHAR(20), Subcode VARCHAR(10), SSID VARCHAR(10), ...	0 row(s) affected	0.078 sec
15	19:01:37	CREATE TABLE EMPLOYEE (SSN VARCHAR(20) PRIMARY KEY, Name VARCHAR(50), Address V...	0 row(s) affected	0.078 sec
16	19:01:42	CREATE TABLE DEPARTMENT (Dno INT PRIMARY KEY, Dname VARCHAR(50), Mgrssn VARCHA...	0 row(s) affected	0.078 sec
17	19:01:48	CREATE TABLE DLOCATION (Dno INT, Dloc VARCHAR(50), PRIMARY KEY (Dno, Dloc), FOREI...	0 row(s) affected	0.047 sec
18	19:01:52	CREATE TABLE PROJECT (PNo INT PRIMARY KEY, PName VARCHAR(50), PLocation VARCHAR(...	0 row(s) affected	0.079 sec
19	19:01:57	CREATE TABLE WORKS_ON (SSN VARCHAR(20), PNo INT, Hours INT, PRIMARY KEY (SSN, P...	0 row(s) affected	0.063 sec
20	19:02:15	CREATE TABLE CUSTOMER (Cust_id INT PRIMARY KEY, cust_name VARCHAR(50))	0 row(s) affected	0.047 sec
21	19:02:20	CREATE TABLE ITEM (item_id INT PRIMARY KEY, item_name VARCHAR(50), price INT NOT NULL)	0 row(s) affected	0.031 sec
22	19:02:25	CREATE TABLE SALE (bill_no INT PRIMARY KEY, bill_date DATE, cust_id INT, item_id INT, qt...	0 row(s) affected	0.094 sec
23	19:02:42	ALTER TABLE STUDENT ADD Email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.125 sec
24	19:03:03	ALTER TABLE EMPLOYEE MODIFY Salary INT NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
25	19:03:07	ALTER TABLE ITEM MODIFY price INT NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
26	19:03:23	ALTER TABLE STUDENT RENAME COLUMN SName TO StudentName	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.046 sec
27	19:03:42	DROP TABLE DLOCATION	0 row(s) affected	0.031 sec
28	19:04:22	INSERT INTO CUSTOMER VALUES (1,'Vanshika'), (2,'Roshani'), (3,'Sneha'), (4,'Priya'), (5,'Karan'), (6,'Meena'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
29	19:04:36	INSERT INTO ITEM VALUES (101,'Pen',20), (102,'Notebook',50), (103,'Bag',800), (104,'Shoes',1200), (105,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
30	19:04:50	INSERT INTO SALE VALUES (1,CURDATE(),1,101,5), (2,CURDATE(),2,104,1), (3,CURDATE(),3,105,2), (4,'2...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
31	19:05:06	INSERT INTO SALE VALUES (1,CURDATE(),1,101,5), (2,CURDATE(),2,104,1), (3,CURDATE(),3,105,2), (4,'2...	Error Code: 1062. Duplicate entry '1' for key 'sale.PRIMARY'	0.016 sec
32	19:05:17	SELECT s.bill_no, c.cust_name, i.item_name FROM SALE s JOIN CUSTOMER c ON s.cust_id = c.Cust_id JOI...	3 row(s) returned	0.000 sec / 0.000 sec

30	19:04:50	INSERT INTO SALE VALUES (1,CURDATE(),1,101,5), (2,CURDATE(),2,104,1), (3,CURDATE(),3,105,2), (4,'2...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
31	19:05:06	INSERT INTO SALE VALUES (1,CURDATE(),1,101,5), (2,CURDATE(),2,104,1), (3,CURDATE(),3,105,2), (4,'2...	Error Code: 1062. Duplicate entry '1' for key 'sale.PRIMARY'	0.016 sec
32	19:05:17	SELECT s.bill_no, c.cust_name, i.item_name FROM SALE s JOIN CUSTOMER c ON s.cust_id = c.Cust_id JOI...	3 row(s) returned	0.000 sec / 0.000 sec

33	19:06:26	SELECT s.bill_no, (i.price * s.qty_sold) AS Total_Amount FROM SALE s JOIN ITEM i ON s.item_id = i.item_id L...	10 row(s) returned	0.000 sec / 0.000 sec
34	19:07:19	SELECT DISTINCT c.cust_name FROM SALE s JOIN CUSTOMER c ON s.cust_id = c.Cust_id JOIN ITEM i O...	7 row(s) returned	0.000 sec / 0.000 sec
35	19:07:57	SELECT c.cust_name, COUNT(s.item_id) AS Total_Products FROM SALE s JOIN CUSTOMER c ON s.cust_i...	10 row(s) returned	0.031 sec / 0.000 sec
36	19:10:31	SELECT i.item_name FROM SALE s JOIN ITEM i ON s.item_id = i.item_id WHERE s.cust_id = 5 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
37	19:11:19	CREATE VIEW Bill_View AS SELECT s.bill_no, s.bill_date, s.cust_id, s.item_id, i.price, s.qty_sold, (i.pr...	0 row(s) affected	0.031 sec
38	19:11:45	CREATE VIEW Weekly_Sales AS SELECT bill_date, SUM(i.price * s.qty_sold) AS Daily_Total FROM SALE s J...	0 row(s) affected	0.031 sec

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